

PRCP-1001 – Rice Leaf Disease using CNN

1. Introduction & Problem Statement

Rice is a staple food crop in Asia, and its productivity is often reduced by leaf diseases. Timely identification and classification of rice leaf diseases is essential for effective crop management and improved yield.

The aim of this project is to build a **Convolutional Neural Network (CNN) model** that can classify rice leaf images into three major categories of diseases:

- **Bacterial Leaf Blight**
- **Brown Spot**
- **Leaf Smut**

2. Dataset Description & Preprocessing

- The dataset used contains **119 JPG images** of rice leaves.
- Class distribution:
 - Bacterial Leaf Blight → 40 images
 - Brown Spot → 40 images
 - Leaf Smut → 39 images

Images were preprocessed by:

- Resizing all inputs to **$224 \times 224 \times 3$ (RGB)**.
- Normalizing pixel values by dividing by 255.
- Converting labels to categorical (one-hot encoding).
- Splitting dataset into training (95 images) and testing (24 images).

3. Exploratory Data Analysis (EDA)

- A bar chart of class distribution confirmed balanced representation across all three classes.
- Sample visualizations were plotted to verify the dataset contained correctly labeled disease images.

4. Model Building

A **baseline CNN architecture** was implemented with the following layers:

1. **Convolutional Layers** with ReLU activation for feature extraction.
2. **MaxPooling Layers** to reduce dimensionality.
3. **Flatten Layer** to convert feature maps into a dense vector.
4. **Dense Layers** for classification.
5. **Softmax Output Layer** for multi-class prediction.

The model was compiled using:

- Optimizer: **Adam** (initially)
- Loss Function: **Categorical Crossentropy**
- Metrics: **Accuracy**

Training

- Epochs: **30**
- Batch Size: **32**
- Validation Split: **20% of training set**

Baseline Model Performance

- Test Accuracy: **~79%** with Adam optimizer.

5. Data Augmentation Experiments

To improve generalization, **ImageDataGenerator** was used with transformations such as:

- Rotation (45°)
- Width & Height shift
- Shear & Zoom transformations
- Horizontal flip

This expanded dataset variability and improved model robustness.

6. Model Comparison with Different Optimizers

The CNN model was trained with **4 different optimizers** after augmentation:

Optimizer Test Accuracy

Adam **79%**

RMSProp 70%

SGD 41%

Adagrad 58%

- Adam optimizer performed best, providing the highest accuracy.
- SGD performed poorly, showing difficulties in convergence.

7. Prediction & Visualization

- Predictions were made on the test dataset.
- Visualization of **100 random test samples** showed correct predictions for all samples.
- This confirmed strong performance of the CNN with Adam optimizer.

8. Challenges & Risks

- **Small dataset size (119 images)** limits model generalization.
- **High computational time** required due to CNN training.
- **Optimizer sensitivity:** large variation in accuracy across optimizers.
- **Data Loading/Processing** can be error-prone if not handled properly.

9. Conclusion

- A CNN model was successfully developed for rice leaf disease detection.
- Among the optimizers tested, **Adam achieved the highest accuracy (79%)**.
- The model generalized well on unseen test data, making it suitable for preliminary disease detection tasks.

10. Future Work

- Expanding the dataset with more images for improved generalization.
- Testing advanced architectures like **Transfer Learning (ResNet, VGG, Inception)**.
- Deploying the model in mobile/edge devices for real-time farmer assistance.